Further results from the cycle-space machinery

Research note prepared by Codex and collaborating agents

10 November 2026

Abstract

We develop several consequences of the November 10 manuscript. The general fixed-gap path bound improves from order $b/\log^2 b$ to order $b/\log b$. For two-terminal series-parallel networks, a sharp additive estimate gives length at least $b-O_\varepsilon(1)$. A small-cut inverse theorem describes pair correlations and conditional concentration near the one-half bound. For random graphs at the Alon–Krivelevich density, the pancyclicity excess above $\log_2 n$ has order $\ln n/\ln\ln n$. The lower coefficient is $2/3-o(1)$ and the upper coefficient is $2+o(1)$. The upper proof refines the published construction and corrects several intermediate statements. For four-regular multigraphs, we prove a doubled-edge reduction and verify the exact conjectured bound through nine vertices. The general linear path and exact four-regular questions remain unresolved here.

Scope and verification status

The source is *The number of cycles in a graph and the minimum number of edges in a pancyclic graph*, dated November 10, 2026. Its local source is `paper/cycle-count/pancyclic-paper.tex` at repository revision `a965788`. We use its definitions of rank, outside parts, and forest probability $p(G,W)$. Rank means $\beta(G)=|E(G)|-|V(G)|+\operatorname{comp}(G)$. An even subgraph is an edge set of even degree at every vertex. A cycle is connected and two-regular.

This note gives mathematical proofs and reproducible finite checks. Its new results have not been formalized in Lean. Existing formalized theorems and the main manuscripts were not changed. The finite checks support the stated identities and constructions. They do not establish the asymptotic theorems. Publication priority for the refinements has not been established.

Direction	Result in this note
Path stability	$\Omega_\varepsilon(b/\log b)$ in general; $b-O_\varepsilon(1)$ for terminal series-parallel networks.
Near-half structure	Quantitative two-link pair mass and concentration, given many disjoint bounded-cut cyclic regions.
Sparse random hosts	Matching order $\ln n/\ln\ln n$ for the second-order excess at the stated p_*.
Four-regular graphs	A smaller remaining class; exact doubled-edge deficit; verification through order nine.
Other targets	An edge-frequency corollary and explicit limits of the present arguments.

1 Path stability and the size of a marked subgraph

These results use the forest proof in the November 10 manuscript. The parameter extension below is a mathematical argument, not a Lean formalization. It improves the general path bound to order $b/\log b$. The linear bound remains open for general graphs. For two-terminal series-parallel networks, an elementary argument gives a stronger additive bound.

All logarithms in this section have base two, except for ln and exp. Write $P_{uv}(N)$ for the number of simple paths between distinct vertices u, v. Write $b = \beta(N)$.

1.1 From paths to forest outside parts

Lemma 1.1. *Let N be a connected simple graph, and let A be any u–v path of length ℓ. Add a new vertex w and edges uw, wv to obtain H. Let $W = A \cup \{uw, wv\}$. Then*

$$\beta(H) = b + 1, \qquad \beta(W) = 1, \qquad |E(W)| = \ell + 2, \qquad P_{uv}(N) \le p(H, W)2^b.$$

Proof. Each terminal path P gives the cycle $P \cup \{uw, wv\}$ of H. Its outside part is $P \setminus E(A)$, which is a forest. Suppose that $P \setminus E(A) = Q \setminus E(A)$. Then $P \triangle Q$ is an even edge set supported on the forest A. It must be empty, so $P = Q$. There are $2^{\beta(H)-\beta(W)} = 2^b$ outside parts. Their forest proportion is $p(H, W)$. □

The formalized forest theorem gives $\ell > \sqrt{b+1} - 2$ for fixed positive path-density gap and sufficiently large b. The manuscript's size extension gives $\ell > (b+1)/\log^2(b+1) - 2$. The next argument removes one logarithmic factor for a fixed gap.

1.2 A fixed-gap forest estimate at size $r/\log r$

Theorem 1.2. *Fix $0 < \delta < 1/2$, and set*

$$A_\delta = 6\exp(16/\delta), \qquad a_\delta = \frac{1}{12A_\delta}.$$

For sufficiently large r, the following statement holds. Let G be a connected simple graph of rank r. Let W be a nonempty union of cycles satisfying

$$|E(W)| \le a_\delta \frac{r}{\log r}, \qquad \operatorname{comp}(W) \le \frac{\log r}{100}.$$

Then $p(G, W) \le 1/2 + \delta$.

Proof. We use the manuscript's reduction, cut lemmas, spectral estimate, and second-moment argument. The details below check each estimate affected by the new parameters. Constants in $O_\delta(\cdot)$ may depend on δ.

Put $x = \log r$, $y = \log x$, $D = \lfloor x/10 \rfloor$, and $s = \lfloor (D-1)/2 \rfloor$. Set $L = \lfloor A_\delta x \rfloor$. The reduction gives a subcubic multigraph Γ and a marked vertex set P. Every vertex outside P has degree three, $|P| \le 2|E(W)|$, and $\Gamma[P]$ has at most $x/100$ components. Also $\beta(\Gamma) = r$ and $2r - 2 \le |V(\Gamma)| \le 2r + 2|E(W)|$. Let $\mathcal{F}$ mean that the random even subgraph is a forest on $\Gamma - P$. It suffices to rule out $\mathbb{P}(\mathcal{F}) > 1/2 + \delta$.

Assume this strict inequality. Choose a maximal vertex-disjoint family $\mathcal{Q}$ of cycles of length at most D in $\Gamma - P$. The small-cut lemma gives

$$|\mathcal{Q}| < \frac{(D+1)2^{D-1}}{\delta} = O_\delta(xr^{1/10}).$$

For their vertex union S, this implies

$$|S| = O_\delta(x^2 r^{1/10}) = o(r/x).$$

Let $J = \Gamma - (P \cup S)$, and put $v = |V(J)|$. This graph is simple and has girth greater than D. The vertex and deficit estimates become

$$v = 2r + O_\delta(r/x), \qquad d_J := \sum_{z \in V(J)} (3 - \deg_J z) \le 3(|P| + |S|) \le \frac{6a_\delta r}{x} + o(r/x).$$

Consequently,

$$\frac{Ld_J}{2v} \le \frac{3}{2} A_\delta a_\delta + o(1) = \frac{1}{8} + o(1).$$

Let B be the non-backtracking matrix of J. For even j with $6x \le j \le L$, the spectral estimate gives

$$2^{-j} \operatorname{tr} B^j \ge \left(1 - \frac{d_J}{2v}\right)^j - 3v2^{-j/2} \ge 1 - \frac{Ld_J}{2v} - 3vr^{-3} \ge \frac{1}{2}$$

for sufficiently large r. For the cycles C of J of length at most L, put $\mu = \sum_C 2^{-|C|}$. The repeated-vertex walk estimate from the manuscript has error at most

$$\frac{3}{2} L^2 2^{-s} \mu = o(\mu).$$

Therefore

$$(1 + o(1))\mu \ge \frac{1}{2} \sum_{\substack{6x \le j \le L \\ j \text{ even}}} \frac{1}{2j} = \frac{1}{8} \ln(A_\delta/6) + o(1).$$

In particular, $\mu \ge \frac{1}{16} \ln(A_\delta/6) = 1/\delta$ for sufficiently large r.

It remains to check the pair correlations. Use $t_0 = \lfloor y/2 \rfloor$ in the manuscript's link argument. For disjoint cycles C, C' of length at most L, it gives

$$\ell(C, C') \le \operatorname{comp}(\Gamma[P]) + \frac{2L}{t_0} + \frac{(t_0 + 1)2^{t_0 - 1}}{\delta} + \left(\frac{8L}{D}\right)^2 \le \frac{x}{100} + o(x).$$

Here the three added terms are respectively $O_\delta(x/y)$, $O_\delta(y\sqrt{x})$, and $O_\delta(1)$. The exceptional-partner estimate thus remains

$$2^{x/100 + o(x)} L^3 2^{-s} \le 2^{x/100 - x/20 + o(x)} = o(1).$$

The same weighted variable Z therefore satisfies

$$\mathbb{E}Z = \mu, \qquad \mathbb{E}Z^2 \le 2\mu^2 + (1 + o(1))\mu.$$

Since $\mathcal{F}$ implies $Z = 0$, the second-moment lemma yields

$$\mathbb{P}(\mathcal{F}) \le \frac{1}{2} + \frac{1 + o(1)}{4\mu} \le \frac{1}{2} + \frac{\delta}{2}$$

for sufficiently large r. This contradicts the assumption. □

Corollary 1.3. *Fix $0 < \varepsilon \le 1/2$. For sufficiently large b, if*

$$P_{uv}(N) \ge (1/2 + \varepsilon)2^b,$$

then every terminal path has length

$$\ell > a_{\varepsilon/2} \frac{b + 1}{\log(b + 1)} - 2.$$

In particular, $\ell = \Omega_\varepsilon(b/\log b)$.

Proof. Apply Lemma 1.1. If the displayed length inequality failed, Theorem 1.2 would give $P_{uv}(N) \le (1/2 + \varepsilon/2)2^b$, a contradiction. The one-component condition holds for sufficiently large b. □

The threshold is not explicit because the supporting asymptotic estimates are not explicit. The additive -2 comes from the added two-edge path. Neither issue affects the asymptotic order.

Corollary 1.4. *Suppose $r \to \infty$ along connected simple graphs G and nonempty unions of cycles W. If $|E(W)| = o(r/\log r)$ and $\mathrm{comp}(W) \le (\log r)/100$, then $p(G, W) \le 1/2 + o(1)$.*

Proof. For each fixed $0 < \delta < 1/2$, the size hypothesis eventually meets Theorem 1.2. Thus $\limsup p(G, W) \le 1/2$. □

1.3 An additive theorem for two-terminal series-parallel networks

A two-terminal series-parallel network starts with one edge and its two endpoints. Series composition identifies one terminal from each network. Parallel composition identifies the corresponding terminal pairs. Apart from those identifications, the two networks are disjoint. Parallel edges are permitted and are counted separately. This definition imposes a condition on the terminals, not only on the underlying graph.

Theorem 1.5. *Let (N, u, v) be a two-terminal series-parallel network of rank b and terminal distance ℓ. Then*

$$P_{uv}(N) \le 2^{b-1} + 2^{\ell-1}.$$

Consequently, for $0 < \varepsilon \le 1/2$,

$$P_{uv}(N) \ge (1/2 + \varepsilon)2^b \quad \Longrightarrow \quad \ell \ge b - \log \frac{1}{2\varepsilon} \quad \Longrightarrow \quad \ell \ge \frac{b}{\log(1/\varepsilon)}.$$

Proof. Every terminal path is a join, so $P_{uv}(N) \le 2^b$ for every connected network. Define $h(t) = (1 + 2^{-t})/2$ for $t \ge 0$. We prove the stronger normalized statement

$$\frac{P_{uv}(N)}{2^b} \le h((b - \ell)_+), \qquad (z)_+ = \max\{z, 0\}.$$

It holds for a single edge.

For series composition, write b_i, ℓ_i, P_i for the two input networks. Then $b = b_1 + b_2$, $\ell = \ell_1 + \ell_2$, and $P = P_1 P_2$. For $s, t \ge 0$,

$$h(s + t) - h(s)h(t) = \frac{(1 - 2^{-s})(1 - 2^{-t})}{4} \ge 0.$$

Also $(b_1 - \ell_1)_+ + (b_2 - \ell_2)_+ \ge (b - \ell)_+$, and h is decreasing. The normalized statement follows by multiplication.

For parallel composition, $b = b_1 + b_2 + 1$, $\ell = \min\{\ell_1, \ell_2\} \ge 1$, and $P = P_1 + P_2$. Since $b_1, b_2 \ge 0$,

$$P \le 2^{b_1} + 2^{b_2} \le 2^{b_1 + b_2} + 1 = 2^{b-1} + 1 \le 2^{b-1} + 2^{\ell-1}.$$

Together with $P \le 2^b$, this is the normalized statement. Induction proves the theorem's first inequality.

The density hypothesis now gives $\varepsilon 2^b \le 2^{\ell-1}$. Thus $b \le \ell + \log(1/(2\varepsilon))$. Since $\ell \ge 1$, the right side is at most $\ell \log(1/\varepsilon)$. □

The coefficient is sharp when $\varepsilon = 2^{-b}$. Take a series of $b-1$ triangles between u, v, and add the edge uv. For $b \geq 3$, the result is simple, has rank b, distance one, and $2^{b-1}+1$ terminal paths. For $b = 2$, a direct edge in parallel with two internally disjoint two-edge paths has the same count. For $b = 1$, use a triangle with adjacent terminals.

The theorem fails for arbitrary choices of terminals in a series-parallel graph. On vertices $0, 1, 2, 3, 4$, take

$$E = \{01, 02, 03, 04, 12, 14, 23\}, \qquad u = 3, \quad v = 4.$$

This graph has rank three and terminal distance two. Its seven terminal paths are

$$3014, \quad 30214, \quad 304, \quad 32014, \quad 3204, \quad 32104, \quad 3214.$$

Thus $P = 7 > 2^{3-1} + 2^{2-1} = 6$. The graph is a triangle with two vertices successively attached to existing edges, so it has no K_4 minor. It does not satisfy the two-terminal composition hypothesis for terminals $3, 4$.

1.4 An elementary structural reduction

Proposition 1.6. *Let N be connected, with distinct terminals u, v. Let N_0 be the union of the blocks on the u–v route in the block-cut tree. Count bridges as blocks. Put $s = \beta(N) - \beta(N_0)$. Then*

$$P_{uv}(N) = P_{uv}(N_0) \leq 2^{\beta(N_0)} = 2^{b-s}.$$

In particular, $P_{uv}(N) > 2^{b-1}$ implies $s = 0$.

Proof. A simple terminal path cannot enter a block outside the terminal route. It would have to revisit the articulation vertex where it left that route. Thus all terminal paths lie in N_0. Apply the join bound to N_0. Ranks add across blocks, so s is a nonnegative integer. □

This removes cyclic blocks outside the terminal route from the unresolved problem. It does not control the internal structure of a single block. The fixed-gap argument still loses a factor $\log b$ there. Its spectral estimate counts walks of length at least order $\log b$. A marked set of order b creates a constant degree deficit, whose effect accumulates along those walks. The present retuning does not prove general linear stability.

Finite checks. The accompanying script checks 1,728 distinct composition states for networks with at most 12 edges. It verifies the inequality with exact integers across 27,264 series or parallel transitions. It also enumerates the seven counterexample paths and the sharp simple examples for ranks three through nine. These checks do not verify the asymptotic estimates or replace their proofs.

2 A quantitative inverse for the small-cut estimate

This section refines the small-cut argument from the source manuscript. It gives both a pair-structure conclusion and a concentration conclusion. It does not give a decomposition of every graph with many cycles.

Let G be a connected multigraph. Let $A_1, \ldots, A_k$ be disjoint cyclic regions, where $k \geq 2$ and $|\delta A_i| \leq d$. Set $\Delta = \max\{d-2, 0\}$ and $K = 1 + \Delta$. For a uniform even subgraph ξ, define

$$E_i = \{\xi \cap \delta A_i = \varnothing, \quad \xi \cap E(G[A_i]) \neq \varnothing\}, \qquad p_i = \Pr(E_i), \quad Z = \sum_i \mathbf{1}_{E_i}, \quad \mu = \sum_i p_i.$$

Let ℓ_{ij} count the links between A_i and A_j. The source manuscript gives

$$\Pr(E_i \cap E_j) = 2^{\ell_{ij}-1} p_i p_j.$$

Every region is proper, since $k \geq 2$. The cut formula therefore improves the elementary lower bound to

$$p_i = 2^{\mathrm{comp}(G-A_i)-|\delta A_i|}(1 - 2^{-\beta(G[A_i])}) \geq 2^{-d}, \qquad \mu \geq k2^{-d}. \tag{1}$$

Theorem 2.1. *Suppose that* $0 \leq \eta < 1/2$ *and*

$$f := \Pr\big(\xi \cap E(G[A_i]) \text{ is a forest for every } i\big) \geq \tfrac{1}{2} - \eta.$$

Define the ordered one-link mass by

$$D_1 = \frac{1}{\mu^2} \sum_{i \neq j:\, \ell_{ij}=1} p_i p_j.$$

Then

$$D_1 \leq \frac{4\eta}{1+2\eta} + \frac{K}{\mu} \leq \frac{4\eta}{1+2\eta} + \frac{K2^d}{k}. \tag{2}$$

Choose I, J *independently with* $\Pr(I = i) = \Pr(J = i) = p_i/\mu$. *Then*

$$\Pr(I \neq J,\ \ell_{IJ} = 2) \geq 1 - \frac{4\eta}{1+2\eta} - \frac{2K}{\mu}. \tag{3}$$

Writing $\nu = \Pr(Z > 0)$, *we also have*

$$\frac{1}{2 + K/\mu} \leq \nu \leq \tfrac{1}{2} + \eta, \tag{4}$$

$$\frac{\mathrm{Var}(Z \mid Z > 0)}{(\mu/\nu)^2} \leq 2\eta + (\tfrac{1}{2} + \eta)\frac{K}{\mu}. \tag{5}$$

Proof. The exceptional-partner lemma bounds each region's partners with at least three links by Δ. For these pairs, use $\Pr(E_i \cap E_j) \leq p_i$. For one-link pairs, the product moment is $p_i p_j$. For two-link pairs, it is $2p_i p_j$. Together with the diagonal terms, this gives

$$\mathbb{E}Z^2 \leq 2\mu^2 - D_1\mu^2 + K\mu. \tag{6}$$

On the event defining f, every E_i fails. Thus $\Pr(Z = 0) \geq f$, and $\nu \leq 1/2 + \eta$. Cauchy–Schwarz gives $\mathbb{E}Z^2 \geq \mu^2/\nu$. Substitution in (6) proves (2).

The weighted diagonal mass is at most $1/\mu$, since $p_i \leq 1$. The weighted mass of exceptional ordered pairs is at most Δ/μ. Connectedness gives $\ell_{ij} \geq 1$ for distinct regions. Subtract these masses and (2) from one to obtain (3).

Dropping D_1 from (6) gives the lower bound in (4). Finally, $\mathbb{E}(Z \mid Z > 0) = \mu/\nu$, and

$$\frac{\mathrm{Var}(Z \mid Z > 0)}{(\mu/\nu)^2} = \frac{\nu\mathbb{E}Z^2}{\mu^2} - 1 \leq 2\nu - 1 + \nu K/\mu.$$

The upper bound on ν proves (5). □

For fixed d, let $k \to \infty$ and $\eta \to 0$. The theorem forces almost all weighted pair mass onto distinct two-link pairs. It also forces Z/μ to concentrate near two, conditionally on being positive. That conditioning event has probability tending to one half. These conclusions describe the near-equality mechanism of this particular second-moment argument.

Transfer from a cycle count. The chord construction gives $c(G)/2^r \le p(G^+, W) + 2^{-t}$ when t edges are selected. Hence $c(G) \ge (1/2 - \eta)2^r$ implies $p(G^+, W) \ge 1/2 - \eta - 2^{-t}$. Suppose that the regions above are disjoint from $V(W)$ in G^+. The forest event outside W implies the event defining f. The theorem therefore applies with $\eta + 2^{-t}$, when that number is less than one half. A separate argument must produce many regions with bounded cuts.

A more general second-moment boundary. If a proposed improvement uses only $\mathbb{E}Z = \mu$ and $\mathbb{E}Z^2 \le 2\mu^2 + K\mu$, it cannot force an error exponentially small in μ. For $K > 0$, take $Z = 0$ with probability $(\mu + K)/(2\mu + K)$ and $Z = 2\mu + K$ otherwise. Both moment conditions hold with equality, and

$$\Pr(Z = 0) - \tfrac{1}{2} = \frac{K}{2(2\mu + K)}.$$

This is a limitation of those moment inequalities alone. It is not a graph construction or an obstruction to stronger graph estimates.

3 The second-order scale in a sparse random host

Status. The lower bound below has a complete first-moment proof. The upper bound is a parameter refinement of the published Alon–Krivelevich construction. Its proof uses their expansion and Hamilton-path inputs. Neither result is formalised in Lean. We have not established publication priority for these refinements.

All logarithms in this section are natural, except $\log_2$. Put $D = np$. For a pancyclic graph G on n vertices, define

$$\mathrm{Pex}(G) = \min\{|E(H)| - n : H \subseteq G \text{ is spanning and pancyclic}\}.$$

If G is not pancyclic, set this quantity to $+\infty$.

Theorem 3.1 (Second-order bounds in the Alon–Krivelevich range)**.** *Define*

$$p_1 = p_5 = \frac{2\ln\ln n}{n}, \qquad p_2 = p_3 = \frac{50\ln n}{n\ln\ln n}, \qquad p_4 = \frac{\ln n + 10\sqrt{\ln n}}{n}, \qquad p_* = 1 - \prod_{i=1}^{5}(1 - p_i).$$

For $G \sim G(n, p_*)$, with probability tending to one,

$$\log_2 n + \left(\frac{2}{3} - o(1)\right)\frac{\ln n}{\ln\ln n} \;\le\; \mathrm{Pex}(G) \;\le\; \log_2 n + (2 + o(1))\frac{\ln n}{\ln\ln n}.$$

Consequently,

$$\mathrm{Pex}(G) - \log_2 n = \Theta\left(\frac{\ln n}{\ln\ln n}\right) \qquad \textit{with high probability.}$$

The upper bound also holds for every $p \ge p_*$. The lower bound holds whenever $np \sim \ln n$, including values arbitrarily close above the pancyclicity threshold.

The last lower-bound assertion does not supply a finite upper bound below p_*. Thus this theorem does not settle the full-threshold question.

3.1 A uniform count of small subgraphs

Lemma 3.2 (Kernel count)**.** *Let $G \sim G(n, D/n)$, with $D \geq 2$. Let $S \geq 2$ be an integer, and let $R > 0$. The probability that G contains a connected subgraph F satisfying*

$$\delta(F) \geq 2, \qquad q := |E(F)| - |V(F)| \geq 1, \qquad |E(F)| \leq \min\{S, Rq\}$$

is at most

$$S \sum_{q=1}^{S} \left(\frac{(2S)^{10} D^R}{n} \right)^q.$$

Proof. Write $e = |E(F)|$. Suppress each maximal path whose internal vertices have degree two. The resulting connected multigraph has t vertices and $t+q$ edges. Every vertex has degree at least three. The degree identity gives $1 \leq t \leq 2q$, and hence $t + q \leq 3q$.

Label the kernel vertices and order its edges. There are at most $(2q)^{6q}$ lists of edge endpoints. There are at most $(S+1)^{3q}$ lists of positive path lengths. There are at most n^{e-q} assignments of distinct host vertices to a fixed subdivision. These bounds can count invalid or repeated descriptions, which only increases the upper bound.

Each described simple subgraph has occurrence probability $(D/n)^e$. After summing over t, its expected count is at most

$$2q(2q)^{6q}(S+1)^{3q} n^{-q} D^e \leq (2S)^{10q} n^{-q} D^{Rq}.$$

There are at most S possible values of e for each q. Sum these bounds and apply Markov's inequality. □

Lemma 3.3 (Independence of a small family of short cycles)**.** *Let $G \sim G(n, D/n)$, with $D \geq 2$, and put*

$$L = \left\lfloor \frac{2(\ln n - 40 \ln\ln n)}{3 \ln D} \right\rfloor.$$

Assume $L \geq 3$. With probability $1 - O((\ln n)^{-17})$, every family of at most L distinct cycles, each of length at most L, is linearly independent over $\mathbb{F}_2$.

Proof. Set $S = \lceil (\ln n)^2 \rceil$ and $R = 3L/2$. For sufficiently large n,

$$D^R \leq \frac{n}{(\ln n)^{40}}, \qquad (2S)^{10} \leq (\ln n)^{21}.$$

Lemma 3.2 shows that the probability of a forbidden subgraph is

$$O\Big((\ln n)^2 (\ln n)^{-19}\Big) = O((\ln n)^{-17}).$$

Suppose a family of at most L short cycles is dependent. Choose a minimal dependent subfamily $C_1, \ldots, C_k$, and let F be its union. Then $3 \leq k \leq L$. Minimality gives $\dim\langle C_1, \ldots, C_k \rangle = k - 1$. Each edge occurs an even positive number of times in the family. Thus, with $e = |E(F)|$,

$$e \leq \frac{1}{2} \sum_{i=1}^{k} |C_i| \leq \frac{kL}{2} \leq S.$$

The union is connected. Otherwise, each component would give a proper dependent subfamily. Also, $\delta(F) \geq 2$ and

$$q = e - |V(F)| = \beta(F) - 1 \geq k - 2 \geq 1.$$

Since $k \le 3(k-2)$ for $k \ge 3$,

$$e \le \frac{kL}{2} \le \frac{3L}{2}(k-2) \le Rq.$$

This contradicts the absence of a forbidden subgraph. □

The lemma concerns families of at most L cycles. It does not assert independence of the entire family of short cycles. This distinction avoids an unjustified bound on a minimal dependency's size.

3.2 The improved lower bound

Proposition 3.4 (Finite lower bound)**.** *Under the hypotheses of Lemma 3.3, with the same probability, every spanning pancyclic subgraph H of G satisfies*

$$|E(H)| - n \ge \log_2(n-2) + L - 3 - \log_2\left(\frac{L(L+1)}{2} - 3\right).$$

In particular, if $D \sim \ln n$, then with high probability

$$\mathrm{Pex}(G) \ge \log_2 n + \frac{2\ln n}{3\ln D} - O(\ln\ln n).$$

Proof. Choose in H one cycle of each length $3, \dots, L$. Let W be their union, and write $r = \beta(H)$, $t = \beta(W)$, and $M = |E(W)|$. Lemma 3.3 gives

$$t \ge L - 2, \qquad M \le \frac{L(L+1)}{2} - 3.$$

The outside-part length count gives

$$n - 2 \le M 2^{r-t}.$$

The Hamilton cycle makes H connected, so $r = |E(H)| - n + 1$. Taking binary logarithms proves the finite bound. For $D \sim \ln n$, the rounding and the $40\ln\ln n$ margin change L by only $O(1)$. Also, $\log_2 M = O(\ln\ln n)$. □

The denominator $\ln D$ matters for a quantitative error term. The assumption $D \sim \ln n$ permits replacement by $\ln\ln n$ with a relative $o(1)$ error. It does not always permit replacement with only an additive $O(\ln\ln n)$ error.

The coefficient $2/3$ improves the disjoint-cycle coefficient $1/2$ in the proposed research list. The proof requires independence, rather than vertex-disjointness. A theta graph explains the first obstruction to this argument. Its three cycles are dependent, and their total length is twice its number of edges. At lengths near $\frac{2}{3}\log_D n$, its occurrence threshold becomes relevant. This observation limits the independence method, not the true pancyclicity excess.

3.3 A two-digit refinement of the upper construction

The primary source is Alon and Krivelevich, *Sparse Pancyclic Subgraphs of Random Graphs*, SIAM Journal on Discrete Mathematics 39 (2025), 562–574, doi:10.1137/23M1598969. We checked the authors' 29 November 2024 manuscript. Their Theorem 1 gives the threshold p_* used above. The following proof changes the short-length encoding parameters in Sections 3–5.

Lemma 3.5 (Corrected robust expansion statement)**.** *Put* $\beta = 2(\ln\ln n)^2/\ln n$ *and* $q_0 = 50\ln n/(n\ln\ln n)$*. Let* $L_0 = \Theta(n/\sqrt{\ln n})$*. With high probability,* $J \sim G(n, q_0)$ *has the following property. Every set* U *of size at least* $n - 2L_0$ *contains a set* U' *such that*

$$|U'| > |U| - \beta n, \qquad |N_{J[U']}(A)| \geq \frac{|A|}{3\beta} \quad (0 < |A| \leq \beta n).$$

Proof. Let $s = \lceil \beta n \rceil$. With high probability, every two disjoint s-vertex sets have an edge between them. Indeed, the failure probability is at most

$$\binom{n}{s}^2 e^{-q_0 s^2} \leq \exp(s\,(2\ln(en/s) - q_0 s)) = o(1),$$

because $q_0 s \sim 100\ln\ln n$ and $\ln(en/s) \sim \ln\ln n$. Assume this property holds.

Start with U and repeatedly delete a nonempty set A that violates the required expansion. Each deleted set has at most βn vertices. Suppose the accumulated deleted set B first reaches size at least βn. Then $\beta n \leq |B| < 2\beta n$. In the original induced graph $J[U]$, its external neighbourhood has size less than

$$\frac{|B|}{3\beta} < \frac{2n}{3}.$$

Thus $U \setminus (B \cup N_{J[U]}(B))$ has at least $n/3 - 2L_0 - 2\beta n$ vertices. For large n, this number exceeds s. Choosing s vertices there and s vertices in B contradicts the assumed edge property. Therefore fewer than βn vertices are deleted before the procedure stops. The remaining set is the required U'. □

This corrects the size assertion in the source's Claim 4.3. Its printed bound $|U'| \geq (1-\beta)n$ can exceed $|U|$ when $|U| = n - 2L_0$. The corrected bound gives $|U'| = n - o(n)$, which suffices for every exposure estimate below. It also supplies the same rooted-tree embedding conditions. For $k = \lfloor \beta n/2 \rfloor$, take maximum degree $\lfloor 1/(4\beta) \rfloor$ and tree size at most $\lfloor n/50 \rfloor$. For $|A| \leq k$, the expansion surplus is at least $|A|/(12\beta) > 1$. For $k < |A| \leq 2k$, that surplus exceeds $n/50$ for sufficiently large n. These are the two hypotheses of the source's Theorem 2.2.

Proposition 3.6 (Refined upper bound)**.** *Let* $p \geq p_*$*, with* p_* *as in Theorem 3.1. Then, with high probability,*

$$\mathrm{Pex}(G(n,p)) \leq \log_2 n + (2 + o(1))\frac{\ln n}{\ln\ln n}.$$

Proof. Use the five independent exposures from the cited construction. In G_2, first expose only edges with both endpoints outside the initial graph H_1. In G_3, first expose only edges with both endpoints outside H_2. The two-set estimate in Lemma 3.5 applies within either reservoir. All path endpoints lie outside its corresponding reservoir. Thus endpoint edges remain independent until their designated queries. Keep its parameters

$$\beta = \frac{2(\ln\ln n)^2}{\ln n}, \qquad K_0 = \left\lfloor \log_2 \frac{\ln n}{6\ln\ln\ln n} \right\rfloor, \qquad K = \left\lfloor \log_2 \frac{n}{\sqrt{\ln n}} \right\rfloor, \qquad L_0 = 2^{K+1} - 1.$$

For integral branching, put

$$a = \left\lfloor (5\beta)^{-1} \right\rfloor, \qquad d = \lfloor \log_a(n/200) \rfloor.$$

Then $d = (1 + o(1))\ln n/\ln\ln n$. The change from real to integral branching has no effect on this estimate.

Replace the short-length parameters by

$$t = 2, \qquad b = \lceil \sqrt{2 \ln n} \rceil.$$

The short base cycle has length $2b + 1$. Attach internally disjoint paths of lengths

$$d + 2 + j \quad \text{and} \quad d + 2 + jb \qquad (0 \le j < b)$$

to its $2b$ designated edges, as in the cited construction. Choosing one path from each group produces every length in

$$[T + 1, T + b^2], \qquad T = 2(d + b + 1).$$

Indeed, each integer $0 \le k < b^2$ has the form $k = j_0 + bj_1$. The resulting cycle has length $T + 1 + k$.

Here are the estimates needed to transfer the source's existence proof.

Initial cycles. The new length $2b+1$ is $o(\ln n / \ln \ln \ln n)$. It is therefore below the longest initial cycle used in Lemma 4.1. That lemma still supplies the required vertex-disjoint initial cycles.

Vertex budgets. The short gadget uses at most

$$O(bd + b^3) = O((\ln n)^{3/2})$$

vertices. This is $o(L_0)$. The binary gadget, connectors, and short gadget together use at most $2L_0$ vertices. Thus Lemma 3.5 applies at every step. Each remaining expander contains $n - o(n)$ vertices.

Path embeddings. Each additional stem has length at most $b(b - 1) = O(\ln n)$. The rooted trees in Lemma 4.5 still have fewer than $n/50$ vertices. Their leaf sets have at least $\beta n/40$ vertices because

$$a^d > \frac{n}{200a} \ge \frac{\beta n}{40}.$$

Process the short paths in their order around the base cycle. For the new endpoint, expose at most $h = \lceil \beta n/100 \rceil$ candidate neighbours in the remaining expander. These edges have not appeared in any earlier exposure. The first-neighbour failure probability is at most

$$\exp(-p_3 h) \le (\ln n)^{-1}.$$

The other endpoint has previously exposed at most h edges. At least $\beta n/40 - h \ge \beta n/80$ leaves therefore have unexposed endpoint edges. The remaining-leaf failure probability is at most

$$\exp(-p_3 \beta n/80) = (\ln n)^{-5/4}.$$

There are $2b = O(\sqrt{\ln n})$ short paths. The sum of their failure probabilities tends to zero. The binary paths and the final connector retain their original estimates. The two-tree embeddings also remain below the size limit. For $a \ge 3$, their two complete trees use at most $3n/200$ vertices together. The connecting stem has $O(L_0) = o(n)$ vertices. Their maximum degree is at most $a + 1 \le \lfloor 1/(4\beta) \rfloor$.

Length coverage. The binary gadget starts at

$$\ell_* = (K_0 + 1)(d + 3) = (1 + o(1)) \log_2 n.$$

There are enough base edges for all designated binary detours and the reserved connector edge. Indeed, with $x = \ln \ln n$,

$$\ln a = x - 2 \ln x - \ln 10 + o(1), \qquad K_0 + 1 > \frac{x - \ln 6 - \ln \ln x}{\ln 2}.$$

These inequalities give $\ell_* - K \to \infty$. Since $b^2 \geq 2\ln n$, the short interval reaches ℓ_* for all sufficiently large n. The remaining interval-covering shortcuts from Lemma 4.8 are unchanged. Only the lengths $3, \ldots, T$ need individual short chords. Here $T = O(\ln n / \ln\ln n)$. Each missing-chord probability is at most $(1-p_5)^n \leq (\ln n)^{-2}$. Their sum tends to zero, as required by the argument of Lemma 4.9.

Hamilton completion. The remaining vertex set still has size at least $n - 2L_0$. Claim 4.7 therefore applies without a parameter change. The number of long interval-covering chords is at most

$$m = \lfloor n2^{-K} \rfloor = O(\sqrt{\ln n}).$$

The final graph has a Hamilton cycle and at most

$$K + bt + m + T + O(1) = K + 2d + 4b + m + O(1) \leq \log_2 n + 2d + O(\sqrt{\ln n})$$

additional edges. Since $\sqrt{\ln n} = o(\ln n / \ln\ln n)$, this proves the claimed upper bound. The conclusion for $p \geq p_*$ follows by monotonicity. □

Further notation corrections in the checked source. These corrections do not require a new probabilistic input. First, the source writes $b^t \geq \ln n > \ell_*$, although $\ell_* \sim \log_2 n$. The choice $b^2 \geq 2\ln n$ gives the required overlap explicitly. Second, Lemma 4.4 states one extra edge in the binary replacement path. For a desired path length $\ell_i = 2^i + 1$, use a connecting stem of length $\ell_i - 2d - 2$. Its positivity follows from $i > K_0$. This gives the required gain 2^i when one cycle edge is replaced. The short-chord argument above covers every length through T. This also supplies the extra t lengths omitted from the printed range in Lemma 4.9.

The exposure budget in Lemma 4.5 also needs correction. The source first permits $n/\ln\ln n$ exposed edges at an endpoint. This exceeds the guaranteed $\beta n/40$ leaves, so its later fresh-leaf subtraction does not follow. The budget $h = \lceil \beta n/100 \rceil$ repairs that step and supports the new union bound.

What remains open in this note. The lower and upper constants do not match. The upper construction still needs $p \geq p_*$. No result here settles the full pancyclicity threshold range. The lower proof uses the outside-part length count, without the stronger forest probability estimate.

4 Reductions for the exact four-regular bound

All graphs in this section are loopless multigraphs. Parallel edges are distinct, and two parallel edges form a cycle. Write $c(G)$ for the number of cycles. The target is

$$c(G) \leq 2^n + n \qquad \text{for every four-regular graph on } n \text{ vertices.} \tag{7}$$

This is Open Problem 2 of Alahmadi, Aldred, dela Cruz, Solé, and Thomassen, *The maximum number of minimal codewords in long codes*, https://arxiv.org/html/1201.6043, Section 6. A focused literature search found no subsequent resolution. This search does not establish that none exists. The results below do not settle (7).

4.1 Complementation and the possible counterexamples

For connected G, its cycle space has 2^{n+1} elements. The pairs $\{X, E(G) \setminus X\}$ number 2^n. Let a count pairs containing two cycles, and let b count pairs containing no cycle. Then

$$c(G) = 2^n + a - b.$$

Both members of a pair counted by a must be Hamilton cycles. Indeed, their degrees sum to four at every vertex. Each cycle contributes at most two there, so both must contribute two. Consequently, a graph without a Hamilton decomposition has $c(G) \leq 2^n$. This is the specialization of the cited paper's Theorem 6.

A graph with a Hamilton decomposition is four-edge-connected. Each Hamilton cycle crosses every nontrivial cut at least twice. For $n \geq 3$, such a graph has edge multiplicity at most two. Each Hamilton cycle uses at most one edge between a given pair of vertices.

4.2 Doubled-edge expansion preserves the proposed bound

Suppose that G has exactly two edges between u and v. Replace these edges by two uw edges and two wv edges, where w is new. Call the resulting graph G'. This operation preserves four-regularity and adds one vertex. Let $a_{uv}(G)$ count cycles using exactly one of the two uv edges.

Proposition 4.1. *For connected four-regular G on $n \geq 3$ vertices,*

$$c(G') = c(G) + a_{uv}(G) + 1, \qquad a_{uv}(G) \leq 2^n.$$

Thus (7) for G implies (7) for G'. Moreover, $a_{uv}(G) = 2^n$ holds if and only if G is the doubled n-cycle.

Proof. Cycles avoiding both uv edges remain unchanged. Each cycle using exactly one gains twice as many choices after expansion. The original two-edge cycle is replaced by two two-edge cycles. This proves the identity.

Let N result from deleting the two uv edges. If N is disconnected, its terminals lie in different components, and $a_{uv}(G) = 0$. Otherwise $\beta(N) = n - 1$, and

$$a_{uv}(G) = 2P_{uv}(N) \leq 2 \cdot 2^{\beta(N)} = 2^n.$$

Here the path bound follows by injecting paths into the affine space of $\{u, v\}$-joins. The argument allows parallel edges.

Suppose equality holds. Every join of N is then a path. For any such path A, deleting $E(A)$ leaves a forest. Otherwise A together with an edge-disjoint cycle would be a join that is not a path. Thus $|E(A)| \geq \beta(N) = n - 1$, and every terminal path is Hamiltonian. Fix one such path. An edge joining nonconsecutive vertices of that path would give a shorter terminal path. Therefore all edges of N join consecutive vertices of the fixed path. The terminal degrees are two, and every internal degree is four. Successive degree equations force every consecutive pair to have multiplicity two. Adding the deleted pair makes G a doubled cycle. The converse follows by counting the 2^{n-1} terminal paths in a doubled path. □

More generally, replace the doubled edge by a doubled path with t new internal vertices. Call this graph G_t, with $G_0 = G$. Let $b_{uv}(G)$ count cycles avoiding both uv edges. Direct counting gives

$$c(G_t) = 2^t a_{uv}(G) + b_{uv}(G) + t + 1, \tag{8}$$

$$\big(2^{n+t} + n + t - c(G_t)\big) = \big(2^n + n - c(G)\big) + (2^t - 1)\big(2^n - a_{uv}(G)\big). \tag{9}$$

If G is not a doubled cycle, then $a_{uv}(G)$ is even and at most $2^n - 2$. Hence each expansion sequence increases the deficit by at least $2(2^t - 1)$. This conclusion assumes only the displayed initial deficit, not the full conjecture.

Corollary 4.2. *If (7) fails, a counterexample of minimum order has all the following properties: it is connected, Hamilton-decomposable, and four-edge-connected. Its underlying simple graph has minimum degree at least three. Its parallel pairs form a matching.*

Proof. Disconnected graphs cannot be minimum counterexamples. Apply the smaller-order bounds to their components and use $\sum_i 2^{n_i} \le 2^{\sum_i n_i}$. Complementation gives the Hamilton decomposition, which gives four-edge-connectivity.

Suppose a vertex w has only two distinct neighbors, u and v. Each Hamilton cycle uses one edge to each neighbor at w. Thus both incident multiplicities are two. For $n > 3$, there is no uv edge. Such an edge would close a triangle in whichever Hamilton cycle contains it. Suppress w and replace its four edges by a doubled uv edge. The resulting graph is loopless, four-regular, connected, and has one fewer vertex. Proposition 4.1 then excludes a minimum counterexample. The smaller base cases are the doubled triangle and the four parallel edges on two vertices. Their counts are 11 and 6.

Every vertex now has at least three distinct neighbors and total degree four. It can therefore meet at most one parallel pair. □

This reduction identifies a concrete remaining class: four-regular graphs whose doubled edges form a matching. It does not justify contracting an arbitrary doubled edge. That operation also changes cycles that use both endpoints without using the doubled edge.

4.3 Equality is not unique

The doubled n-cycle has $2^n + n$ cycles. There is another equality example:

$$c(K_5) = \binom{5}{3} + 3\binom{5}{4} + \frac{(5-1)!}{2} = 10 + 15 + 12 = 37 = 2^5 + 5.$$

Thus an exact equality classification cannot contain doubled cycles alone. Proposition 4.1 gives a limited classification. If an equality graph has a vertex with two distinct neighbors, it must be a doubled cycle. No asymptotic stability classification follows from this observation.

4.4 Finite verification

The script `check_four_regular.py` fixes one Hamilton cycle and enumerates the second up to reversal. Every Hamilton-decomposable graph occurs, possibly more than once. Cycle counts use vertex-simple cyclic orders, weighted by edge multiplicities. For orders through eight, an independent implementation enumerates the binary cycle space and tests connectedness and degrees. The two counts agree for every tested graph. The script also checks 2,050 doubled-path expansions and the equality criterion in Proposition 4.1.

n	3	4	5	6	7	8	9
Second Hamilton cycles	1	3	12	60	360	2520	20160
Maximum cycle count	11	20	37	70	135	264	521

These maxima equal $2^n + n$. Together with complementation, the two-vertex base case, and the component argument, this verifies the bound through order nine. The enumeration covers 23,116 second-cycle orderings. It is reproducible evidence, not a Lean certificate or an infinite-order proof. The data are in `four-regular-results.json`.

An additional experiment tested 52,145 contractions of arbitrary doubled edges. The inequality $c(G) \le c(G/\{u, v\}) + 2^{n-1} + 1$ held in every test. Here the contraction identifies u, v and deletes the two resulting loops. This inequality remains unproved and is not used above.

5 Other directions and the remaining gaps

5.1 An edge-frequency consequence

The following implication was proposed in the supplied research list. It follows from the source manuscript's formalized forest estimate. It is recorded here with its denominator and graph assumptions.

Proposition 5.1. *Let G be connected and simple, with sufficiently large rank r. Suppose an edge e lies on a cycle W of length at most $\sqrt{r}$. Then*

$$c_e(G) \le \left(\frac{1}{4} + \frac{10}{\log_2 \log_2 \log_2 r}\right) 2^r.$$

If also $c(G) \ge (1/2 - \eta)2^r$ with $0 \le \eta < 1/2$, then

$$\frac{c_e(G)}{c(G)} \le \frac{1/4 + 10/(\log_2 \log_2 \log_2 r)}{1/2 - \eta}.$$

Proof. Restriction outside W has fibers of size two, namely $\{x, x \triangle E(W)\}$. Exactly one member of each fiber contains e. A cycle containing e has a forest outside part. Thus $c_e(G) \le p(G, W)2^{r-1}$. Apply the source manuscript's forest theorem and divide by the stated lower bound on $c(G)$. □

A connected simple graph of minimum degree three has at most $2r - 2$ vertices. A breadth-first search gives a cycle of length $O(\log r)$. Hence such a graph has an edge satisfying the proposition for large r. For families with $c(G)/2^r \to 1/2$, this gives an edge in at most $(1/2 + o(1))c(G)$ cycles. This is not the exact edge-half theorem for arbitrary graphs.

At a vertex v of degree $d \ge 4$, the exact local half bound is elementary:

$$\sum_{e \ni v} c_e(G) = 2c_v(G) \le 2c(G).$$

Some incident edge therefore has $c_e(G) \le c(G)/2$. A bridge lies in no cycle. Thus a global counterexample with minimum degree at least three would have to be cubic and bridgeless.

5.2 Why the general error-term questions remain

The extended marked-set range in Section 1 concerns a fixed positive error. It does not supply polynomial decay in the error as r grows. The source proof's weighted cycle supply remains logarithmic in its available length ratio. The moment-only example in Section 2 prevents stronger conclusions from those moments alone.

The chord construction gives

$$\frac{c(G)}{2^r} \le p(G^+, W) + 2^{-t}.$$

Its witness can have t components. The current forest estimate allows only $O(\log r)$ components. Consequently, this direct use of the construction cannot make 2^{-t} exponentially small in r. An exponential excess bound needs an additional argument for the omitted cycles or a different witness.

The fixed-edge-budget problem needs a different improvement as well. Multiplying 2^r by a constant does not improve its exponential base when r is proportional to the edge count. No result here proves the proposed base $(2 + 2\sqrt{2})^{1/5}$. The degree-sequence paper supplying that target is Dvořák, Morrison, Noel, Norin, and Postle, *Bounding the number of cycles in a graph in terms of its degree sequence*, https://arxiv.org/abs/1907.12091.

5.3 A boundary for matroid extensions

Let $M = M^*(K_{r+1})$ be the cographic matroid of the complete graph. Its binary cycle space is the graph's cut space and has dimension r. Every nonempty cut $\delta(S)$ is a bond, since both induced sides are connected. Thus every nonzero vector is a circuit, and M has $2^r - 1$ circuits. The one-half theorem cannot extend to all binary matroids or even all regular matroids. This counterexample was already identified in the supplied research list.

5.4 Targets not resolved by this work

General linear path stability remains open in this note. The exact four-regular bound also remains open beyond the reductions and finite checks. The random-host theorem leaves both the sharp correction constant and the full-threshold upper bound open. No new result here proves monotonicity of minimum pancyclic excess or the proposed edge-pancyclic density bound. The supplied list's distinction between cycle-space counts and obligations through every edge remains essential.

6 Reproducible checks and handoff

The following scripts accompany this note. Run them with Python 3 from the repository root. The JSON files record their completed checks.

- `check_path_stability.py`: terminal series-parallel recurrence states, the terminal-pair counterexample, and sharp examples.
- `check_link_stability.py`: exact rational link identities, moment bounds, and stability inequalities on 180 multigraphs.
- `check_random_host_encoding.py`: explicit cycle construction and rank in the two-digit short-length gadget.
- `check_four_regular.py`: exhaustive Hamilton-pair enumeration through order nine, with independent cycle-space counts through order eight.

The shared repository-root handoff is `RESEARCH-HANDOFF-CODEX-2026-11-10.txt`. It records the active files, proof status, source corrections, and remaining questions for Fable 5.2. Parallel work should use separate files to avoid overwriting these fragments.